PRE-ALGEBRA (7TH) · UNIT 8: LINEAR FUNCTIONS AND GRAPHING

Bike Share Showdown

Graph linear functions to choose the best bike or scooter rental.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your city just launched three ways to get around downtown: a bike share, an electric scooter company, and a monthly pass. Each has a different starting fee and price per minute.

You will graph each option as a linear function, compare them, and write a guide that helps riders choose.



CityBike
$1 to unlock
$0.20 per minute

ZoomScoot
$2 to unlock
$0.15 per minute

Learning Goals

- I can make a table and graph for a linear function.
- I can find the slope and y-intercept.
- I can compare two functions on a graph.

Project Timeline

When	What you will do
Day 1	Tables & Functions
Day 2	Slope & Intercept
Day 3	Compare
Final	Rental Comparison Guide + reflection

Part 1: Tables & Functions

Day 1 · CityBike: $1 to unlock and $0.20 per minute.

Ride Data: minutes vs. cost for 8 CityBike rides

Ride 1	Ride 2	Ride 3	Ride 4
5 min · $2.00	10 min · $3.00	12 min · $3.40	15 min · $4.00
Ride 5	**Ride 6**	**Ride 7**	**Ride 8**
20 min · $5.00	25 min · $6.00	30 min · $7.00	8 min · $2.60

1. Complete a table of cost for 0, 5, 10, 15, and 20 minutes.

   Starter: *cost = 1 + 0.20 × minutes*

2. Write a rule for the cost y after x minutes.

3. Is cost a function of minutes? Explain.

4. Graph the CityBike rule.

Part 2: Slope & Intercept

Day 2 · Use the rules.

5. What is the slope of the CityBike line? What does it mean?

6. What is the y-intercept? What does it mean?

7. ZoomScoot costs $2 to unlock and $0.15 per minute. Write its rule.

8. Find the slope between (10, 3.50) and (20, 5) for ZoomScoot.

Part 3: Compare

Day 3 · Graph both rules on the same axes.

9. Which is cheaper for a 10-minute ride? For a 30-minute ride?

10. At how many minutes do they cost the same? Use your graph or a table.

11. Write two sentences for your guide telling riders which to choose.

Final Product: Rental Comparison Guide

Your final product must include:

- Tables and rules for both companies
- Both lines graphed on one set of axes
- Slope and y-intercept meanings
- The break-even point
- Rider guide

Reflection

R1. What does a steeper line mean in this situation?

R2. Why do the two lines cross?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Linear functions, slopes, intercepts, systems, and inequalities are modeled and used correctly to compare options. Goes beyond the minimum with a creative, realistic choice.	Linear functions, slopes, intercepts, systems, and inequalities are modeled and used correctly to compare options.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Rider Guide includes graphs, rules, intersection points, and a clear decision chart. Polished, organized, and visually engaging.	Rider Guide includes graphs, rules, intersection points, and a clear decision chart. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.